

Shashank S

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EXPERIENCE

Senior Machine Learning Engineer, Walmart

Dec 2024 – Jul 2025

Transitioned to full-time graduate study

- Reduced infrastructure cost growth by optimizing PySpark jobs in the adtech ML pipeline: profiled bottlenecks, rewrote shuffle-heavy transformations, and scaled data processing to match ad volume.
- Enabled non-technical teams to self-serve analytics by engineering a Text2SQL agentic system (plain English queries, visualizations, active chat); accelerated data access and decision-making.
- Improved model reliability and deployment speed by building an ML observability framework for production monitoring; standardized ML build, test, and deployment workflows with GitHub Actions.

Machine Learning Engineer III, Glance-InMobi

May 2021 – Nov 2024

Promoted: Data Scientist II → MLE III, Jul 2022

- Delivered personalized content to **100M DAU** at **sub-50ms p99 latency** by building an end-to-end real-time ML pipeline (prediction service, Vertex feature store, Kafka event streams, ELK logging, Airflow retraining).
- Increased engagement by **5%** and conversion by **12%** by developing and iterating recommendation models (multi-armed bandits, Elo ratings, GBMs, graph-based collaborative filtering) for sparse and dense users.
- Boosted time-spent by **+11.68%** overall and **+18.38%** for cold users by co-authoring lookalike modeling research (SVD + K-Means seeding a MAB model).
- Reduced on-call response time and incidents by owning the ML automation platform end-to-end: streaming pipelines, model controllers, automated rollback triggers, and event-driven personalization.
- Explored diverse ML approaches including user-level logistic regression, reinforcement learning with MABs, probability state machines, popularity, and linear models.
- Built automated quiz generation over live video streams in real-time; PoC demonstrated on YouTube Live Streams.
- Drove A/B testing infrastructure and experimentation culture across the org, increasing experiment velocity and data-driven decision-making.

Data Scientist, Bright Money

Jun 2019 – Apr 2021

Promoted: Junior Research Scientist → Data Scientist, Jan 2021

- Founded the data science function at an early-stage fintech; built the ML layer from scratch across the core product, enabling data-driven features and growth.
- Launched *Debt Manager* on Airflow and Django, driving adoption to **10K+ users** and **\$100K+ daily transaction volume** through end-to-end financial analysis and automated payment scheduling.
- Built bill identification, affordability scoring, and overdraft prediction models from zero-to-one, enabling new product capabilities and risk reduction.
- Developed forecasting models for income, expenses, and account balances that powered the product's core retention loop

EDUCATION

Georgia Institute of Technology, Atlanta, USA

May 2026

Master of Science, Computer Science – Specialization: High Performance Computing; GPA 3.87/4.0

- Research: Weight Drift Correction in Analog Computing with *Prof. Hyesoon Kim* – paper in preparation (targeting MICRO 2026)
- Relevant Coursework: GPU Hardware & Software, Scientific Machine Learning, High Performance Computing, High Perf Parallel Computing, HPC Architecture, Deep Learning, Compiler Design, Designing for Curiosity

Sir M. Visvesvaraya Institute of Technology, Bengaluru, India

May 2018

Bachelor of Engineering, Computer Science & Engineering – First Class with Distinction, 79.8%

- Thesis: Image Regeneration with Generative Models
- Electives: Pattern Recognition, Clouds & Clusters, Artificial Intelligence

RESEARCH AND PROJECTS

Weight Drift Correction in Analog Computing

- Algorithmic approaches to correcting drift errors in analog MVM crossbar arrays
- Invertible matrix scaling, SVD-based time-invariant decomposition, and RL-based active Correction; evaluated end-to-end on ResNet18/ImageNet under SONOS and PCMDrift models

GhostGame.io

- Browser-based online multiplayer word game; users can create and join rooms to play with friends.
- Built on Firebase/GCP with Firestore and Cloud Functions.

Automated Debt Management

- USPTO: [US20220261886A1](#), [US20230075411A1](#)
- R&D contributor to the systems underlying these patents

Deep Learning Segmentation and Quantification of Meibomian Glands

- Published in [Biomedical Signal Processing and Control](#)
- CNN-based segmentation strategy for ophthalmic imaging

Image Regeneration with Generative Models

- Use CapsNet as a discriminator in Generative Adversarial Network and demonstrate its application using semantic inpainting on MNIST and face images

Shank's Sessions

- Hosting podcasts and streaming games

Various Tech Toys

- Assorted side projects, mostly built with Streamlit

TECHNICAL SKILLS

Languages: Python, C/C++, PySpark, SQL, Go, Shell, JavaScript, HTML/CSS

Machine Learning & AI: PyTorch, scikit-learn, XGBoost, LightGBM, Recommendation Systems, LLMs, LangChain, Agentic Workflows, Deep Learning, NLP, MLOps, Feature Stores, A/B Testing

High Performance Computing: CUDA, MPI, OpenMP, GPU Profiling, Distributed Systems, Scientific ML (PINNs)

Infrastructure: Spark, Kafka, Docker, Kubernetes, Airflow, Argo, Vertex AI, Prometheus, Grafana, Neptune, Metabase, Django

Cloud & Tools: AWS, GCP, Azure, PostgreSQL, Git, LaTeX, Streamlit, Linux, OpenGL, InkScape, GIMP, Jekyll

HONORS & EXTRAS

ACM ICPC 2015 – [Honorable Mention](#)

Letter of Commendation, Minister of HRD – [AISSCE Board Exam](#)

Section Leader, Code in Place | Stanford University | [2021](#), [2023](#), [2024](#) – Taught Python to 30+ students per cohort

Certifications:

- [Neural Networks & Deep Learning \(deeplearning.ai\)](#)
- [Machine Learning \(Stanford / Coursera\)](#)
- [Android Developer Nanodegree \(Google / Udacity\)](#)